

Accident Angels --- Platform Overview

Scholar Transport Compliance & Operations Platform

1. What this is

Accident Angels is a single codebase serving **three connected apps** that digitise scholar-transport operations for the Alliance transport association (40 members): driver compliance, parent booking and visibility, daily trip running, dormant payments, safety reporting, waiting-time billing, and driver ratings.

It was built after a January 2026 fatal scholar-transport accident led Gauteng to mandate electronic compliance for operators. The platform was pitched to the association and a pilot was agreed.

App	Who uses it	What they do
Angels Driver	Drivers	Manage compliance docs, run daily trips, see the route map, record waiting time, view ratings received
Angels Parent	Parents	Choose and rate a driver, track trips, view charges, report safety concerns
Admin Portal	Association admin	Verify compliance, review safety reports, manage settings --- never sees payment amounts

2. How the app works

This section walks through the platform the way a real user meets it.

2.1 The driver's journey

1. **Sign in.** The driver opens the Driver app, enters their mobile number, and receives a 6-digit one-time code by SMS (shown on screen in dev). After entering it they get a 30-day session.
2. **Onboard & comply.** A new driver completes onboarding and uploads six mandatory documents: Professional Driving Permit, Police Clearance, Passenger Insurance, Roadworthy Certificate, Vehicle Photos, and Driver's Licence. Each is reviewed by the association admin. A driver only becomes ACTIVE once all six are approved.
3. **See the day's trips.** Each morning and afternoon the system auto-generates a trip from the children assigned to that driver and their schedules, then optimises the stop order. The

driver sees a Morning and an Afternoon run.

4. **Run a trip.** Tapping "Start" opens a live map (Leaflet + OpenStreetMap) with GPS, turn-by-turn routing (OSRM), and the ordered stops. Each stop's marker is colour-coded: a green ring means the family is paid up, a red ring means they are overdue --- so the driver can see at a glance who owes.
5. **Pick up a child (with waiting time).** At a stop the driver taps **Arrived**. The parent is instantly notified, and a 3-minute free grace timer starts on screen. If the child is collected within 3 minutes there is no charge. After 3 minutes, R5 per minute accrues (shown live), and the driver then taps **Confirm pickup** to complete the stop. The charge is written to a ledger for the parent's next invoice.
6. **See ratings.** The driver has a "My Ratings" page showing their average score and each parent comment. They may hide a comment's text (for example a one-off complaint) --- but they can never change the numeric score.

2.2 The parent's journey

1. **Sign in** in the same way (number + one-time code).
2. **Add a child** with home and school addresses (geocoded to coordinates).
3. **Choose a driver.** Instead of a flat list, the parent sees a *recommended shortlist*: nearby drivers who are fully compliant and have capacity, ranked by a blend of proximity and rating. Each card shows the driver's star rating and rough distance.
4. **Track and stay informed.** The parent sees today's trips, gets a notification when their driver arrives, and is told (without other families' details) if their child was not collected.
5. **Rate the driver.** After being assigned a driver the parent can leave a star rating (1--5) and an optional comment, which aggregates into the driver's public score. They also see other parents' (non-hidden) reviews.
6. **See charges.** If waiting-time billing is live, the parent sees any accrued waiting charges and a "pending on next invoice" callout.
7. **Report a concern.** From the driver's profile the parent can file a safety report. It goes only to the admin; the driver is told a report exists but never sees its content.

2.3 The admin's journey

1. **Sign in** to the Admin portal.
2. **Verify compliance.** Review each driver's uploaded documents, approve or reject, and rely on automatic expiry alerts (sent to drivers 30, 14, 7 and 1 day before a document expires).
3. **Handle safety reports.** Read parent-filed reports and move them through Open / Under review / Resolved / Dismissed with notes.
4. **Manage settings**, including the master `PAYMENTS_LIVE` switch that keeps all billing dormant until the association is ready. The admin deliberately never sees payment amounts or bank details.

3. Key flows (step by step)

Authentication. Phone -> `send-otp` (code hashed and stored, 5-min expiry) -> code shown/SMSed -> `verify-otp` checks it once and issues a signed JWT session cookie. Codes are single-use; the client guarantees it submits a code exactly once so a valid code is never accidentally replayed.

Daily trip lifecycle. A scheduled job (or first driver visit) generates Trip + TripStop rows from each child's schedule, geocodes and orders the stops, then the driver moves each stop PENDING -> COMPLETED/MISSED. When the last stop is done the trip auto-completes.

Waiting-time charge. Arrived stamps `arrivedAt`. On Confirm pickup the server computes $\text{ceil}((\text{elapsed} - 180\text{s}) / 60) \times R5$, writes a `WaitingCharge` ledger row (marked live only when `PAYMENTS_LIVE` is true), and notifies the parent. Timestamps are normalised to UTC so the maths is correct on any host.

Rating + recommendation. A parent rating upserts one row per parent-and-driver pair and recomputes the driver's stored average. The recommendation endpoint filters to eligible drivers (active, all six documents approved, seats available) and ranks them by $0.6 \times \text{proximity} + 0.4 \times \text{rating}$, where proximity is measured from the centre of that driver's current pickups to the child's home.

Payments gate. All billing code checks `PAYMENTS_LIVE` (a database setting, not an environment variable). It is currently off: charges are recorded but nothing is taken until the association switches it on.

4. Architecture

Three apps, one deployment. A lightweight proxy rewrites the request path based on the subdomain:

```
driver.accidentangels.co.za -> /driver-app/*
parent.accidentangels.co.za -> /parent-app/*
admin.accidentangels.co.za -> /admin/*
```

(Custom domains still need DNS pointed at Vercel; the platform is live now at `accidentangels.vercel.app` using the `/driver-app`, `/parent-app` and `/admin` paths.)

Request flow. Browser -> proxy (host to path) -> page or API route -> session check and role guard -> Supabase (PostgreSQL) via a service-role client.

Security & privacy. Access is enforced in the API layer. Admins never see payment amounts or bank details; parents see only their own child; drivers see only their own data and their parents' paid/overdue status. Row-level security policies protect the database against any direct, non-server access.

5. Technology

Layer	Choice
Framework	Next.js 16 (App Router) + React 19
Language	TypeScript
Styling	Tailwind CSS v4 + shadcn/ui
Database	Supabase (PostgreSQL)
Auth	One-time SMS code + signed JWT session
Maps	Leaflet + OpenStreetMap + OSRM routing
SMS	Africa's Talking (sandbox for now)
Payments	Paystack / DebiCheck / Capitec VRP (built, dormant)
Email	Resend
Hosting	Vercel
Tests	Vitest (unit) + Playwright (browser)

6. Data model (15 tables)

User, OtpToken, Driver, Parent, Child, Association, ComplianceDocument, Contract, Transaction, Payout, PlatformConfig, Trip, TripStop, Notification, Report, WaitingCharge, DriverRating.

The three newest tables --- WaitingCharge (waiting-time ledger), DriverRating (parent ratings) --- plus rating columns on Driver and arrival/charge columns on TripStop were added for Phases 3-5.

7. What has been built

Foundation (Phases 1--2) --- Done

- OTP auth and three-app routing

- Compliance: six mandatory docs, upload, admin review, expiry alerts
- Children and scheduling, trip generation and route optimisation
- Live trip navigation with GPS and routing
- Privacy controls; driver-switch lock when a parent owes money
- In-app notification bells (driver and parent) and SMS stubs
- Parent safety reporting to the admin
- Payment infrastructure (built but dormant)

Phase 3 --- Waiting Time Charges --- Done

- Manual Arrived button (two-stage Arrived then Confirm pickup)
- 3-minute free grace, then R5/minute (rounded up per minute)
- Accrual to a billing ledger, gated by PAYMENTS_LIVE
- Parent arrival notification; missed-stop notification and reroute
- Live timer and running cost on the driver's screen
- Parent charges page and dashboard callout

Phase 4 --- Route Map paid/unpaid --- Done

- Green (paid) / red (overdue) markers on the driver's route map
- Paying vs overdue summary chips
- Strictly driver-only --- payment status removed from parent and admin views

Phase 5 --- Ratings + Recommendation --- Done

- Parents rate their driver out of five with an optional comment
- Driver can hide a comment's text but never the score
- Anonymised public reviews on the driver's profile
- Recommendation engine ranking eligible drivers by proximity and rating
- Driver "My Ratings" page

Quality. 34 unit tests passing; clean production build (93 routes); full end-to-end run; deployed and smoke-tested live. The login bug ("a correct code was rejected") was traced to a double-submission of the single-use code, fixed on the client, and verified in production across repeated automated runs.

8. What is remaining

Deferred features (not in the confirmed pilot scope yet)

Item	Notes
Admin route-overlap view	Zone overlaps for collective fuel/insurance bargaining
Driver of the Month	Admin nomination, badge, testimonials
Per-child levy (Phase 6)	R1--10 per child per month to the association; needs split payments
Dress-code monitoring	Requested at pitch; not started

Operational steps before pilot go-live

Item	Status
Africa's Talking SMS credentials	Pending (code shown on screen for now)
Paystack live keys	Pending (needed before PAYMENTS_LIVE is turned on)
Resend email key	Pending
Custom subdomains (driver./parent./admin.)	Pending DNS at Vercel
Trevor's "day in the life" shadowing	Pending (informs GPS-based waiting detection)
3--5 pilot driver accounts + association admin	Pending

Known follow-ups

- GPS waiting detection: v1 uses a manual Arrived button; automatic stationary-vehicle detection is planned after the shadowing day.

9. Running and deploying

```

npm run dev           # localhost:3000 (driver app by default)
npm run dev:parent    # parent app
npm run dev:admin     # admin portal
npm run test          # unit tests
npm run build          # production build
vercel --prod --yes   # deploy

```

Database migrations are applied by pasting the relevant `scripts/*.sql` file into the Supabase SQL Editor (latest consolidated file already applied).

10. Status at a glance

Area	State
Auth and three-app routing	Live
Compliance and expiry alerts	Live
Trips and navigation	Live
Privacy controls	Live
Safety reporting	Live
Waiting-time charges (Phase 3)	Live (billing dormant)
Route paid/unpaid map (Phase 4)	Live
Ratings and recommendation (Phase 5)	Live
Per-child levy (Phase 6)	Not started
Real SMS / live payments	Awaiting credentials
Pilot onboarding	Pending